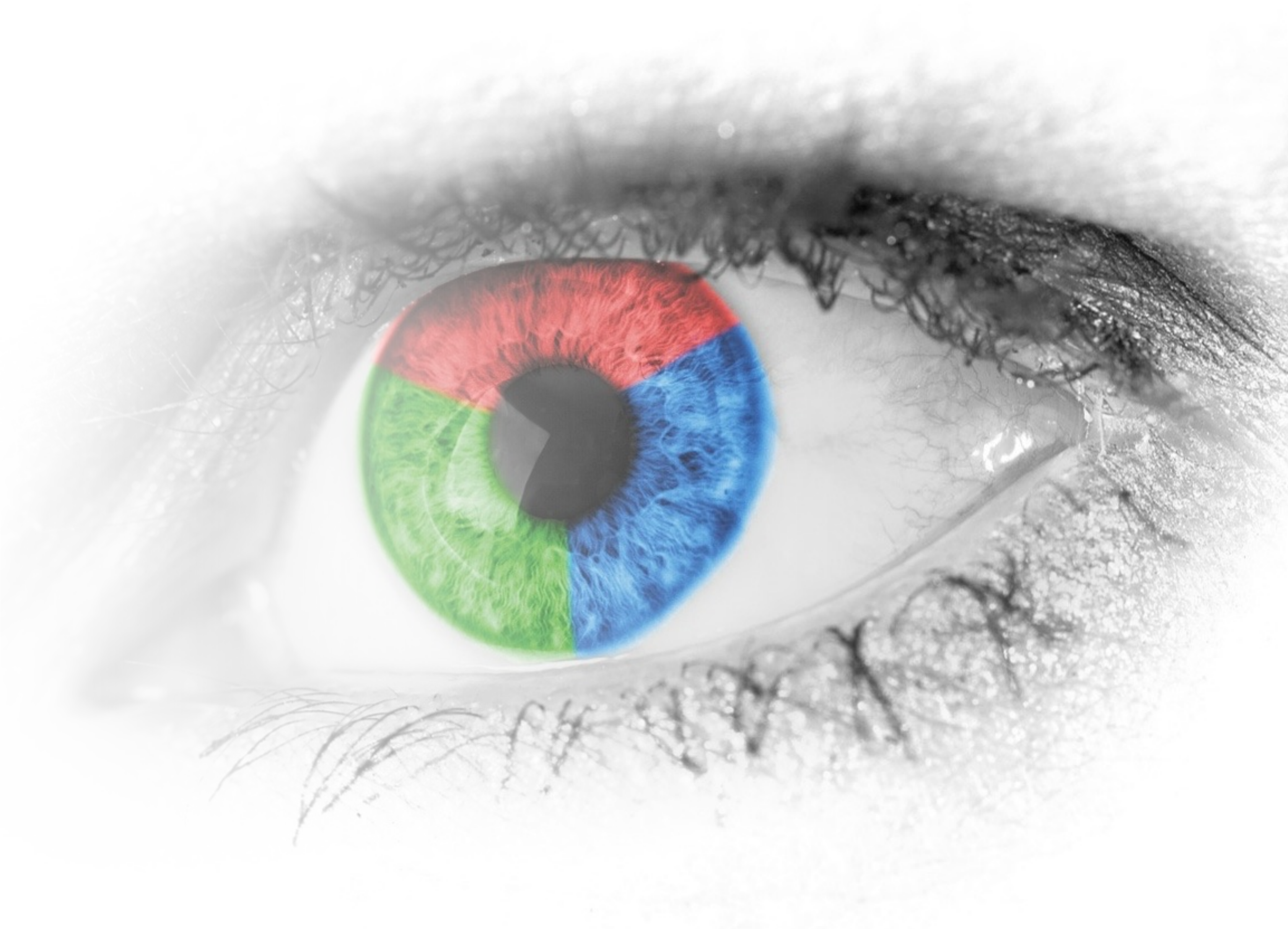
Today’s Agenda

Convolutional Neural Network

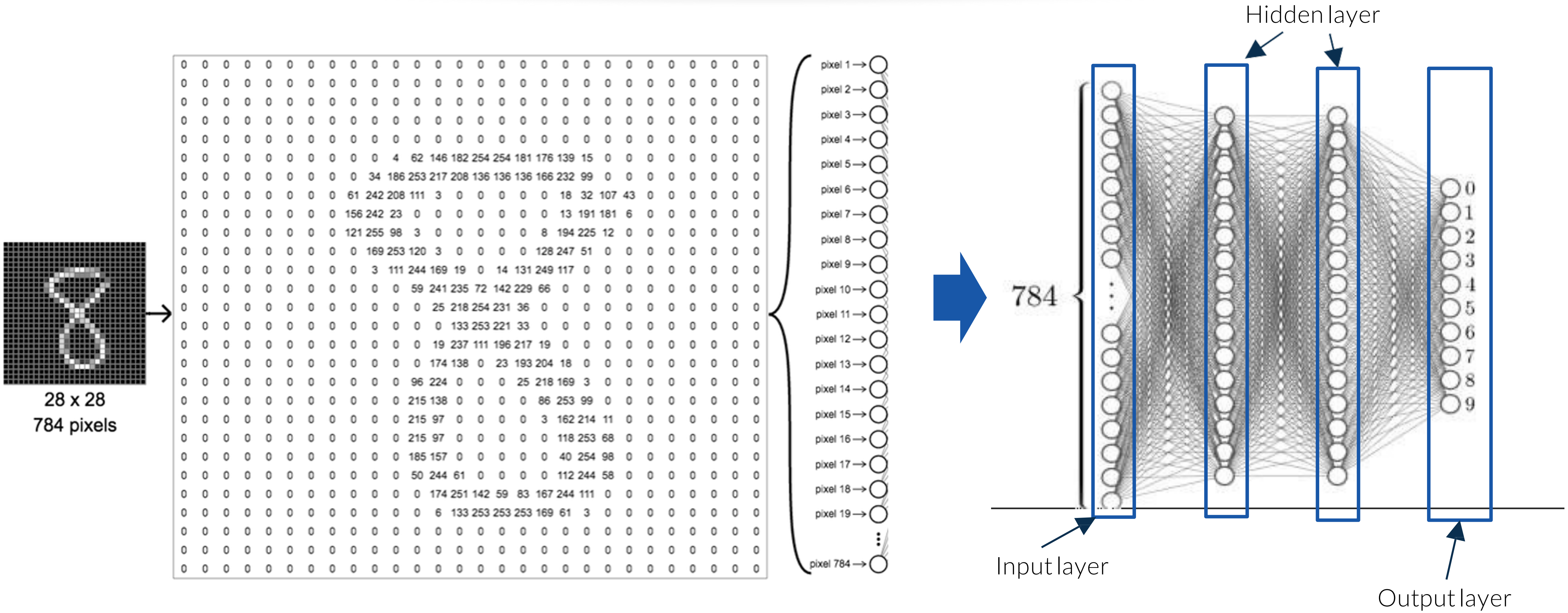
- Problem with ANN
- Why CNN
- Convolution
- Type of layers in CNN
  - Convolution
  - Pooling

Helping computers to see!

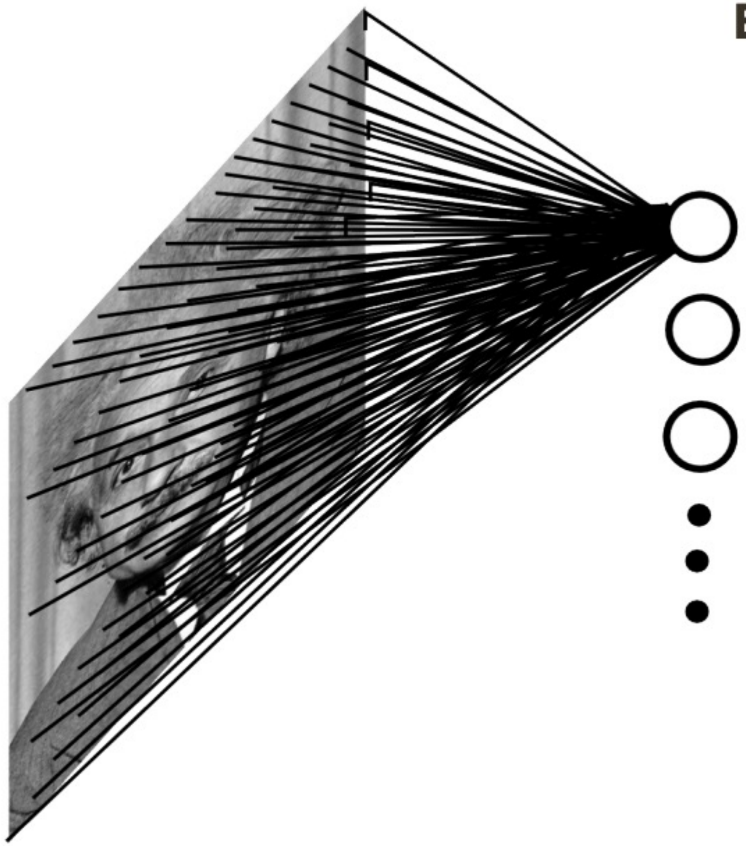


ANN or Multi-Layered Perceptron

Need for CNN: Reason #2



Fully Connected Layer

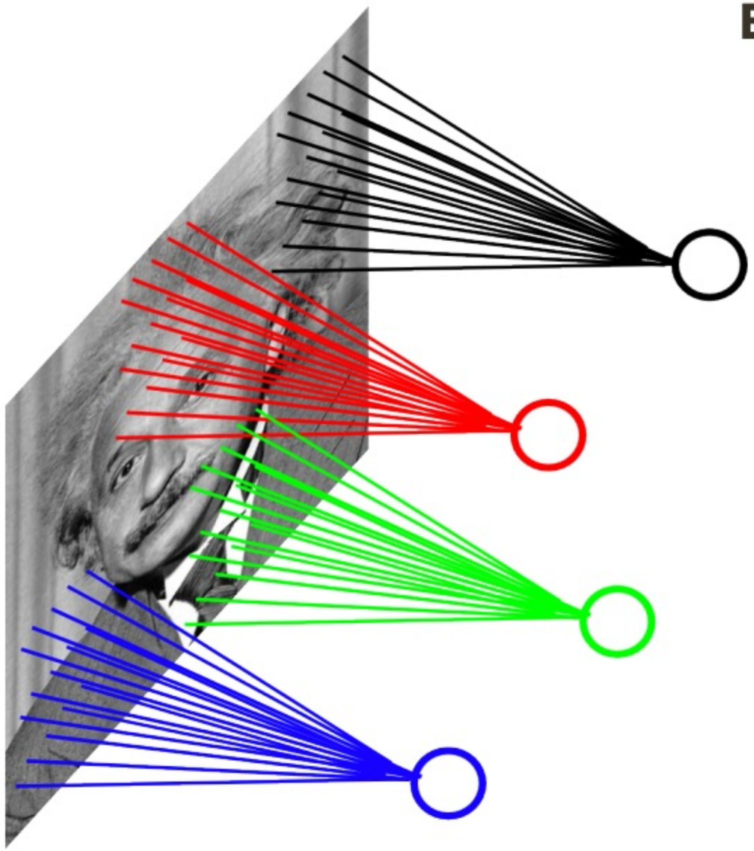


Example: 200 x 200 image (small)
x 40K hidden units

= ~ **2 Billion** parameters (for one layer!)

Slide Credit: Marc'Aurelio Renzato

Locally Connected Layer



Example: 200 x 200 image (small)

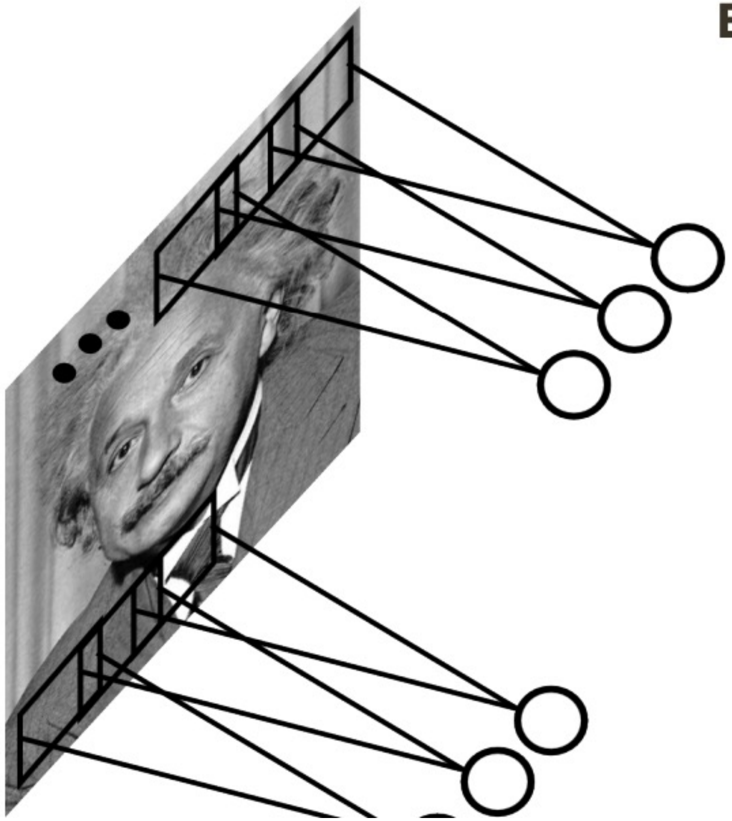
Filter size: 10 x 10

Hidden Units (Neurons): 100

= ~ **4 Million** parameters

Slide Credit: Marc'Aurelio Renzato

Convolutional Layer



Example: 200 x 200 image (small)

Filter size: 10 x 10

= ~ **~~4 Million~~** parameters

= 100+1 parameters

Share the same parameters across the locations (assuming input is stationary)

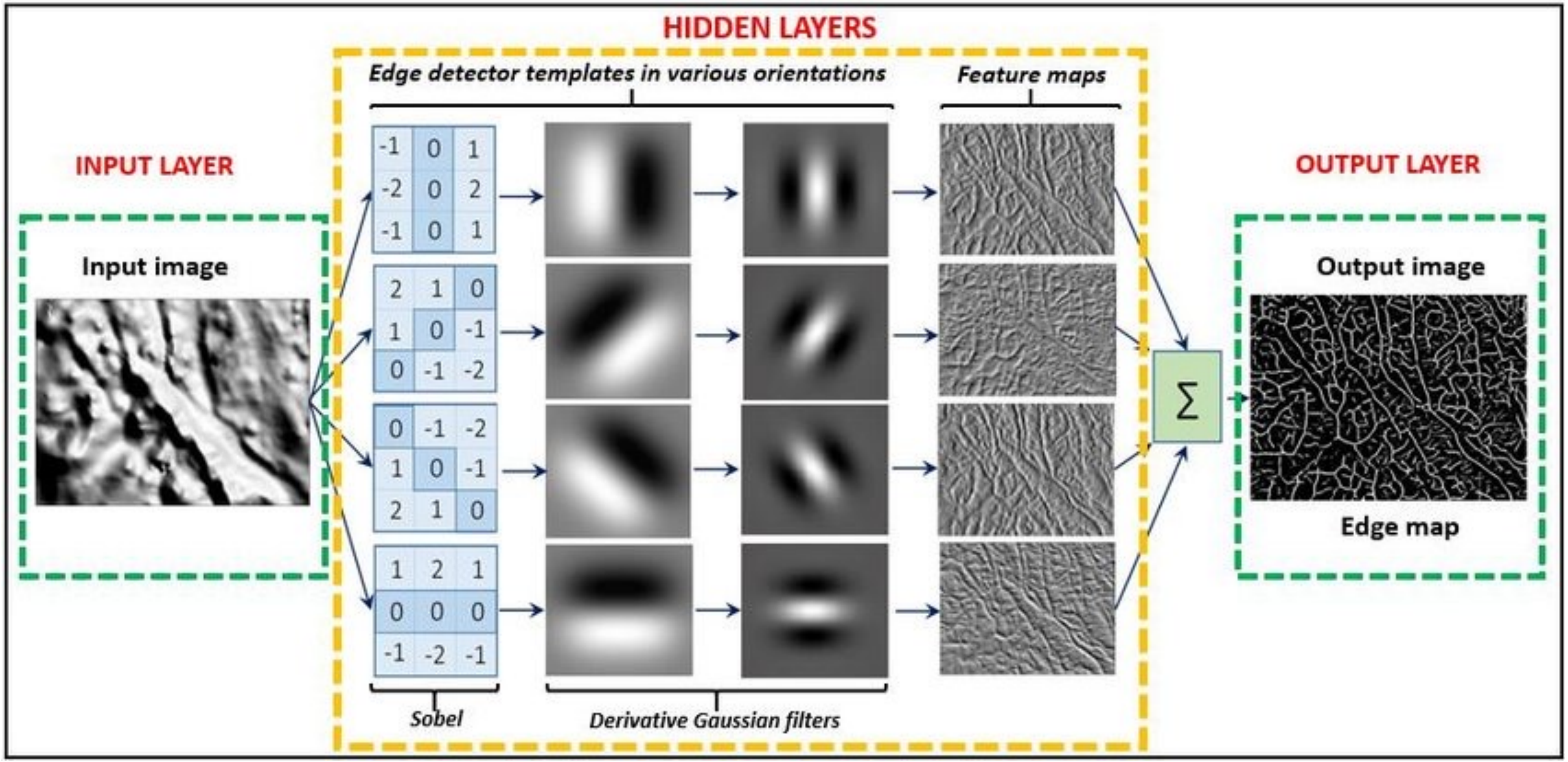
Slide Credit: Marc'Aurelio Renzato

Our Goal Today

Letting the model **learn** relevant visual features to perform the task!

The Convolution Operation

The building blocks of convolutions are essentially dot products over matricies - we multiply the values in a matrix by the corresponding values in another matrix, then add the values together to get a number.



Convolutional Neural Network